



Data Engineer

Tech Interview Titans

Interview Questions & Answers

Experience: 2-5 Year

Coforge Interview Tip (Important)

Interviewers want:

- ✓ Problem-solving approach
- ✓ Real project examples
- ✓ Clear communication
- ✓ Ownership mindset

1. Scenario: Source table has 100M records. How will you load data efficiently?

Answer

I will never load full data every time.

First, I check if the source has a **timestamp or incremental column** like

`last_updated_date`.

I load data in partitions (date-wise or ID ranges) and enable parallel processing.

For large volumes, I prefer **distributed processing (Spark)** instead of traditional SQL jobs.

Why this works:

It reduces load time, cost, and avoids unnecessary reprocessing.

2. Scenario: Late-arriving data is missing from reports. What will you do?

Answer

I implement an **overlap window strategy**.

For example, if daily data loads, I reprocess the last 2–3 days every run.

This ensures delayed records are captured without reprocessing full history.

3. Scenario: Spark job fails with OutOfMemory error.

Answer

First, I identify whether the issue is due to **data skew, large joins, or caching**.

Then I:

- Reduce columns
- Avoid unnecessary cache
- Increase partitions
- Optimize joins (broadcast where applicable)

If required, I tune executor memory.

4. Scenario: Too many small files in Data Lake.

Answer

Small files cause performance issues because Spark spends more time reading metadata.

I fix this by:

- Repartitioning data
 - Merging files during write
 - Running compaction jobs
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5. Scenario: Source system sends duplicate records.

Answer

I identify a **business key** (like order_id).

Then I deduplicate using window functions or Spark `dropDuplicates()`.

Duplicates are logged for audit.

6. Scenario: Business complains about incorrect report numbers.

Answer

First, I validate source vs target counts.

Then I check transformations, filters, and joins.

I trace data lineage step-by-step to find where mismatch happened.

7. Scenario: Schema changes break your pipeline.

Answer

If minor changes, I use schema evolution.

If major changes, I version schemas and coordinate with downstream teams.

Pipelines should never silently fail.



8. Scenario: How do you handle CDC data?

Answer

I use CDC flags like insert/update/delete.

Deletes are handled carefully by marking records inactive instead of hard delete.

9. Scenario: API data ingestion fails randomly.

Answer

I implement retries and timeout handling.

Failures are logged and partial data is rolled back to maintain consistency.

10. Scenario: Query is slow in warehouse.

Answer

I analyze execution plan, indexes, partitions, and joins.

I optimize filters, remove unnecessary columns, and use proper aggregations.

11. Scenario: Data quality issues in production.

Answer

I implement data validation rules like null checks, range checks, and record counts.

Bad records are quarantined, not mixed with clean data.

12. Scenario: Job failed halfway through.

Answer

I design pipelines to be **idempotent**.

Jobs resume from last checkpoint instead of restarting fully.

13. Scenario: How do you monitor pipelines?

Detailed Answer:

I monitor execution time, failures, and volume anomalies.

Alerts are configured to notify immediately.



14. Scenario: Real-time vs batch — how do you choose?

Answer

If business needs instant insights, I use near real-time.
For reporting and analytics, batch is cost-effective.

15. Scenario: Data skew in Spark.

Answer

I identify skewed keys and fix using salting or repartitioning.
This balances workload across executors.

16. Scenario: Handling historical backfill.

Answer

I backfill in chunks to avoid overloading systems.
Historical loads are separated from daily jobs.

17. Scenario: Handling PII data.

Answer

I mask or encrypt sensitive columns.
Access is restricted using roles and policies.

18. Scenario: Pipeline SLA missed.

Answer

I analyze bottlenecks and optimize heavy transformations.
I also parallelize tasks where possible.

19. Scenario: Deployment failure.

Answer

I rollback to previous stable version.
All deployments go through testing and approvals.



20. Scenario: Interview question – Explain your pipeline end-to-end.

Answer

I clearly explain source → ingestion → transformation → validation → storage → reporting.
I also mention error handling and monitoring.

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